Words and Money at ECOSOC

Do the UN's stated priorities match where its money flows?

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Abstract

We ask whether the topics discussed by the UN Economic and Social Council (ECOSOC) align with UN funding patterns, and whether ECOSOC text can help predict changes in UN revenue. We collect 910 ECOSOC summary records (2000 to 2025) and a 244-row panel of UN agency revenue (2011 to 2024). We then build feature sets of increasing richness, including activity and macro controls, theme word-counts, and TF-IDF features. For Q1, we map ECOSOC's agenda using an LDA topic model checked validated against hand labels. For Q2, we predict year-over-year UN revenue growth using five models across nested feature sets and a time-ordered train, validation, and test split. We find partial alignment: ECOSOC discussion is dominated by broad development language, while UN revenue is distributed across agencies with distinct mandates. Adding text features lowers out-of-sample error for some models, especially linear models using TF-IDF features, with the best result from OLS with TF-IDF (test RMSE 0.094). The main limitation is the small yearly sample. To reduce overinterpretation, we use a rolling backtest and emphasize patterns that are stable across feature sets and model specifications.

1. Introduction

Large institutions like the United Nations (UN) often make their stated priorities publicly visible through speeches, resolutions, and published meeting records. However, because these records are numerous and dispersed over time, it is difficult to determine whether funding patterns actually match stated priorities without data science methods. This paper thus asks whether the UN Economic and Social Council's (ECOSOC) stated agenda is reflected in institutional funding patterns.

We focus on ECOSOC here because it coordinates discussion of major economic and social priorities across the UN system, even though it does not directly determine agency budgets. Its meeting records provide a decades-long public text corpus for measuring changes in agenda emphasis over time. We compare these textual patterns with annual UN agency revenue to test whether shifts in discussion correspond to funding outcomes.

We focus on two questions:

1. **Q1 (Alignment).** Do ECOSOC themes line up with UN funding patterns?
2. **Q2 (Prediction).** Do richer text features help predict UN revenue growth?

Our main finding is that ECOSOC's agenda is related to funding patterns, but imperfectly. Development language dominates ECOSOC discussion, while UN revenue is distributed across agencies with different mandates. Text features also improve prediction in some models, especially linear models using TF-IDF features. However, these gains are model-dependent and should be interpreted cautiously because the yearly sample is small.

2. Data

We use three datasets. All the code that collects and cleans them is in the project repository.

2.1 ECOSOC summary records

We collected 910 ECOSOC meeting summary records from 2000 to 2025 from the UN Digital Library. Each record is a PDF that starts with a procedural header (who attended, the agenda, the room and time) and then summarizes what was discussed. We clean out the header text, then aggregate the cleaned text to the year level so it can be aligned with annual funding data. The annual text features include meeting volume, token counts, sentiment, and lexical features for development, humanitarian, climate, gender, conflict, health, and governance language. The total corpus contained around 5.1M words. Coverage varies by year, with sparse coverage in 2000, 2020, and 2025, but most years contain dozens of records.

2.2 UN agency revenue

Using the UN System Chief Executives Board (CEB) financial statistics we built a 244-row table covering 19 agencies from 2011 to 2024[1]. Each observation records revenue from one agency in one year. Total annual revenue increased from about $23.8 billion in 2011 to $43.3 billion in 2024, peaking at around $51.5 billion in 2022. Revenue is concentrated among a small group of large agencies, especially WFP, UNICEF, UNDP, UNHCR, and WHO. We aggregate agency revenues to the annual level to construct total UN revenue, which serves as the outcome variable of the prediction task for Q2.

2.3 Macro controls (WDI)

The macro control variables come from the World Bank Development Indicators (WDI) dataset, which covers 266 countries from 2000 to 2024. We pulled seven indicators: GDP growth, government expenditure as a share of GDP, life expectancy, official development assistance received, the GINI coefficient, FDI inflows, and secondary school enrollment. Of these, four (GDP growth, government expenditure as a share of GDP, life expectancy, and ODA received) are used as controls in the

[1] Note that the number of agencies changed over the period in consideration. In 2011 there were 16 agencies and by 2024 there were 19

supervised baseline. These controls proxy broad global economic and development conditions that may affect UN revenue independently of ECOSOC language. By including them before adding richer text features, we can measure whether Tier 1 lexicon features or Tier 2 TF-IDF features improve prediction beyond a macro-adjusted baseline.

2.4 Summary table

Table 1 summarizes the project's main data sources. The text and revenue data are later aligned at the year level for prediction, which leaves a short annual series and motivates the sample-size caution in Section 5.

Data Source	Rows	Years	Coverage/Scale
ECOSOC records	910 records	2000-2025	Text corpus; ~5.1M words
UN agency revenue	244 agency-years	2011-2024	19 agencies; $485.2B
WDI controls	7 indicators, 4 controls	2000-2024	266 countries, annual global macro controls

Table 1. Data sources used in the project.

3. Methods

3.1 Text preprocessing

We first remove the standard ECOSOC header from each record (room number, attendance list, meeting open time), because that text is repeated but does not reflect substantive agenda content. We then split the text into sentences and group everything by year, which gives us 26 year-level documents. Working at the year level keeps the text lined up with the yearly money data.

3.2 Feature tiers used in the final supervised scope

We build three feature sets that get richer in steps, so we can see how much each layer of text adds:

- **Numeric_only:** basic activity and sentiment features plus the macro controls.
- **Plus Tier 1:** adds theme word-count rates (how often development, health, gender, and other theme words appear).
- **Plus Tier 2:** adds TF-IDF features, which capture specific one- and two-word phrases.

3.3 Unsupervised topic model

To describe the agenda, we fit an LDA topic model to the yearly text. LDA finds groups of words that tend to appear together and calls each group a topic. We tried 6, 10, and 15 topics and picked the number that scored best on a standard coherence measure, which was 6. We then read the top words of each topic and gave it a plain theme label (development, health, climate, and so on). To check the topics are real and not just leftover procedural language, we hand-labeled a 52-record sample by year and measured how often the human label showed up in the model's top three themes for that year. Agreement was about 60%. It was near-perfect for development and health and zero for humanitarian, gender, and conflict, which makes sense because at 6 topics the model only separates a few themes.

3.4 Supervised setup

Target Variable: We estimate whether ECOSOC text features improve out-of-sample prediction of annual UN revenue growth. Let UN_t denote total UN revenue in year t, the dependent variable is defined as: $y_t = log(\frac{UN_t}{UN_{t-1}})$

Thus, model coefficients and prediction errors are interpreted on a year-over-year growth-rate scale.

Model design: To preserve temporal ordering and avoid information leakage, we use a chronological split: training years 2012-2019, validation years 2020-2021, and test years 2022-2024. We compare five supervised regressors: OLS, Ridge, Lasso, Random Forest, and Gradient Boosting. This model set spans unregularized linear, regularized linear, and non-linear tree-based learners. This allows us to test whether any observed gains are model-specific.

We evaluate three nested feature sets:

1. numeric_only: meeting/activity features plus macro controls;
2. +tier1: numeric_only plus lexicon-based thematic rates;
3. +tier2: +tier1 plus TF-IDF text features.

To minimize OVB, we include macro controls from WDI, as listed in the Data section above, in the baseline feature block. These variables proxy broad global economic and development conditions that may influence UN revenues independently of ECOSOC language. Any performance improvement from +tier1 or +tier2 is measured relative to a macro-adjusted baseline.

OLS is the unregularized linear baseline; Ridge (α = 1.0) adds moderate L2 shrinkage to stabilize high-dimensional coefficients; Lasso (α = 0.01, max_iter = 20000) adds L1 shrinkage to encourage sparsity and interpretability. For tree-based models, we use RF (n_estimators = 200, max_depth = 3, random_state = 42) and GBM (n_estimators = 200, max_depth = 3, random_state = 42) with shallow trees to limit overfitting in a small annual sample while preserving non-linear capacity.

Linear models are fit in a `StandardScaler` pipeline so regularization penalties are comparable across differently scaled predictors. Tree models are left unscaled because split-based methods are scale-invariant.

4. Results

4.1 Q1: Alignment

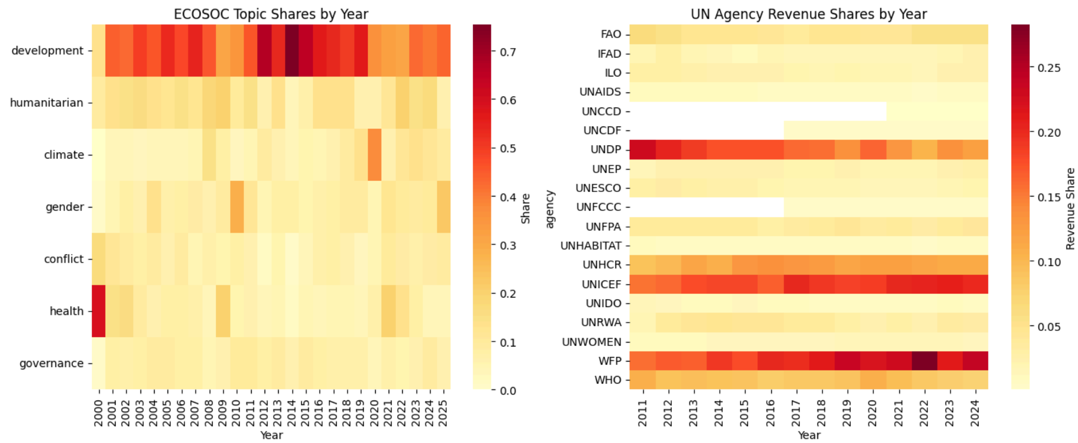

Figure 1. ECOSOC Topic Shares and UN Agency Revenue Shares by Year

Figure 1 compares ECOSOC's agenda with the UN's agency revenue share from 2011 to 2024. Across the whole period, development dominates ECOSOC's discussions, reaching up to 70% of the topic share in 2014. This lines up with the revenue lead of UNDP and UNICEF, which are both development-focused agencies. WFP's revenue grew sharply after 2020, reaching about 25% of the revenue share in 2022, likely driven by COVID-19 food insecurity and the war in Ukraine. But there is no matching jump in humanitarian language in ECOSOC meetings, which suggests that discussion lags behind funding. We also see a spike in health language in 2000, when the AIDS crisis put global health at the center of the debate.

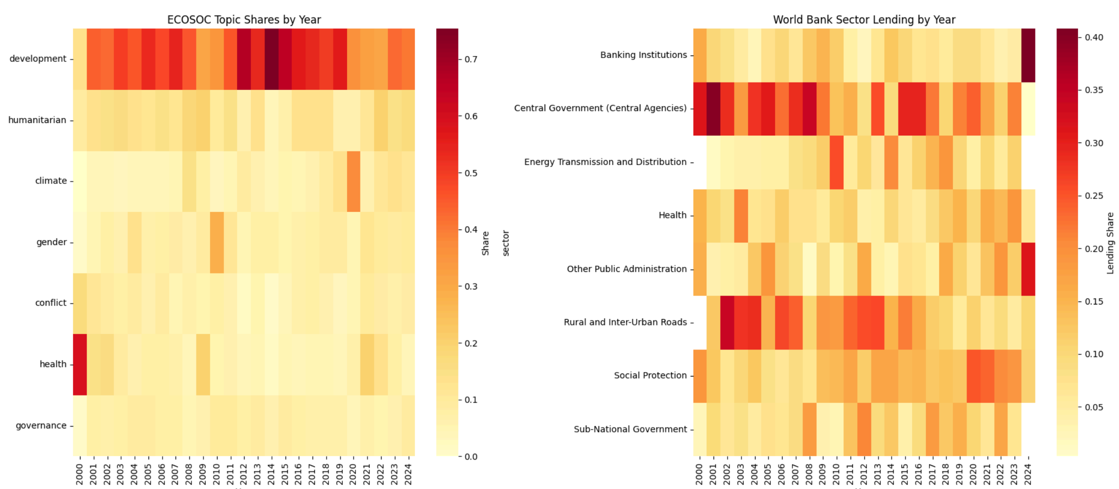

Figure 2. ECOSOC Topic Shares and World Bank Sector Lending Shares by Year

Figure 2 compares ECOSOC's agenda with World Bank sector lending from 2000 to 2024. Even though development and health dominate ECOSOC discussion, World Bank lending stays concentrated in Central Government, which reaches up to 40% of lending in 2001, and in Rural and Inter-Urban Roads. Health lending peaks around 2002 to 2004 and again around 2021 to 2023, which lines up with periods of more health talk in ECOSOC. Banking Institutions lending jumps in 2024 while Central Government lending falls.

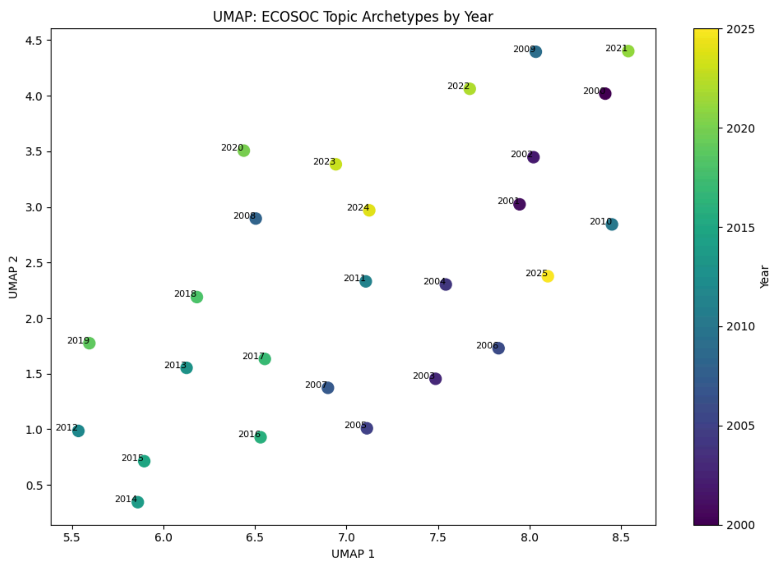

Figure 3. UMAP projection of ECOSOC yearly topic mixtures

Figure 3 is a UMAP projection of ECOSOC's yearly topic mixtures. Each point is a year, and points that sit close together had similar topic patterns. The year 2000 sits alone at the top, which reflects its health-heavy agenda during the AIDS crisis. The early 2000s cluster on the right, the mid-2010s cluster in the bottom left, and recent years from 2019 on are more scattered. That scatter suggests ECOSOC's agenda has become more varied and harder to predict lately.

4.2 Q2: Prediction

We use RMSE as the main metric because it measures out-of-sample error directly in the target's units (log-growth), which is easy to read. We do not lean on R-squared here because it is unstable in our small test set and often goes far negative.

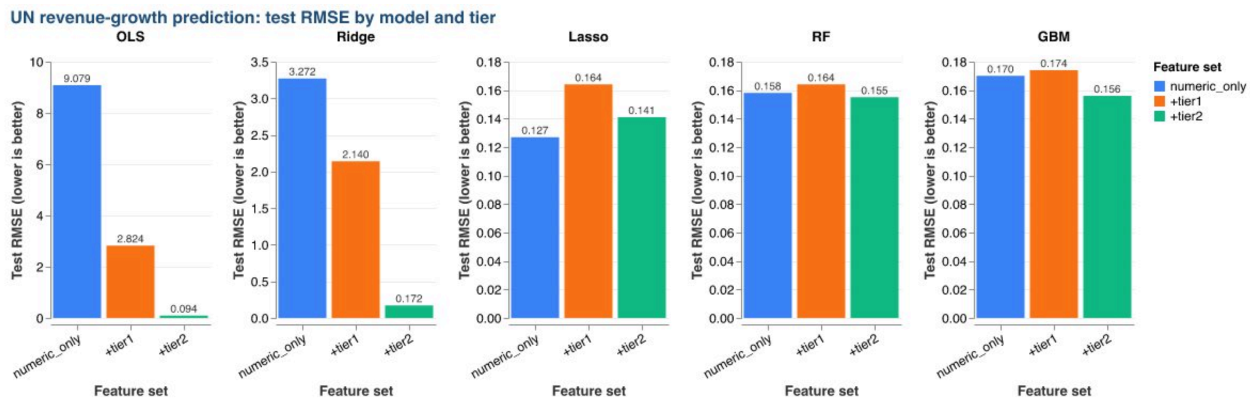

Figure 4. Test RMSE by model and feature tier. Note the model-specific vertical scales. Text features help the linear models most.

Figure 4 reports test RMSE by model and feature tier. The lowest fixed-split error is OLS plus Tier 2 (RMSE 0.094). OLS drops sharply as the text gets richer (9.079 numeric-only, 2.824 plus Tier 1, 0.094 plus Tier 2), and Ridge follows the same path (3.272, 2.140, 0.172). The tree models improve only a little with Tier 2 (Random Forest 0.158 to 0.155; Gradient Boosting 0.170 to 0.156). Lasso is mixed, doing best with numeric-only (0.127) and slightly worse once text is added. Lasso remained sparse, selecting only a few predictors: gdp_growth and oda_received in numeric_only; gdp_growth plus lex_humanitarian, lex_development, and lex_gender in +tier1; and gdp_growth plus four TF-IDF terms in +tier2.

Overall, richer text, especially the Tier 2 TF-IDF features, tends to improve prediction, though the size of the gain depends on the model. The strongest gains show up in the linear models, while the tree models change less but mostly in the same direction. Given the small yearly sample, we read these results as pointing in a clear direction rather than as exact, settled effects.

5. Discussion + Limitations

A few cautions shape how far these results should be pushed.

- **Small sample:** We have only a handful of years of revenue data against thousands of possible text features. That makes in-sample fit easy to inflate and test scores jumpy, which is exactly why the numeric-only linear models post huge errors and the Tier 2 models post very small ones. The rolling backtest and our preference for results that hold across models are our main defenses.
- **Summary text, not transcripts:** ECOSOC publishes summaries, not word-for-word transcripts, so some of what was actually said is smoothed over or dropped. We filter the boilerplate and validate the topics, but the text is still a summary.

- **Association, not cause:** Our models show patterns, not causes. The macro controls reduce confounding from broad global trends, but we do not claim that ECOSOC words cause funding to change.
- **Why controls matter:** Without the macro controls, global booms or downturns could look like a text effect. Including them means any text gain is measured on top of those broad conditions.

A natural next step is to work at the meeting level instead of the year level, which would turn 26 yearly points into hundreds of observations and make the prediction results much steadier.

6. Conclusion

On Q1, ECOSOC's words and the money it coordinates point the same way but do not match in detail: the agenda is dominated by one development theme which consistently occupied over 50% of the topic share throughout the period, while the money is spread across many agencies. On Q2, ECOSOC text does carry real signal about next-year UN revenue growth, strongest in the linear models with TF-IDF features (OLS test RMSE 0.094), though the small sample means we read it as directional. The practical takeaway is that stated priorities and spending rhyme without lining up exactly, and the clearest next step is to push the analysis to the meeting level so the patterns can be pinned down with more data.

References

- UN Digital Library. ECOSOC summary records (E/...). documents.un.org.
- UN System Chief Executives Board for Coordination (CEB). Financial Statistics. unsceb.org.
- World Bank. World Development Indicators (WDI).
- Tools: gensim (LDA), scikit-learn (supervised models), sentence-transformers, VADER.

Appendix A. Member contributions

Member	Contribution
Tahvia Williams	Unsupervised topic modeling (topic_model.ipynb) and topic validation (label_transcripts.ipynb), plus the topic interpretation text. Wrote Sections 3.1, 3.2, 3.3, and the Q1 narrative in 4.1.
Vadim Rudic	Data controls and alignment prep: the WDI controls and the alignment visuals (the two heatmap comparisons and the UMAP), and support on the Q1 interpretation with Tahvia. Wrote Section 2.3, 4.1 the Conclusion.
Ada Meltzer	Corpus and data summary statistics, Table 1, descriptive QA checks, and formatting support. Drafted the Abstract and Introduction, and led Section 2.
Enrico Madani	Supervised pipeline (models.ipynb): the growth target, time-ordered split, the five models across feature sets, ablations, robustness checks, and supervised figures. Wrote Sections 3.4, 4.2, and 5.